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ACCURATE MODEL FOR EFFECTIVE DIELECTRIC CONSTANT OF MICROSTRIP WITH VALIDITY UP TO MILLIMETRE-WAVE FREQUENCIES

Indexing terms: Microwave devices and components, Modelling, Microstrip

A closed-form expression is presented for the effective dielectric constant of single microstrip lines which is valid with high accuracy up to mm-wave frequencies. The formula given has been designed for use in MIC and MMIC CAD programs as well as in the indirect measurement of substrate permittivities by resonance techniques.

Introduction: As pointed out in a recent study by Pucel¹ the last three years have witnessed an intensive revival in the field of monolithic microwave integrated circuits (MMICs). For the economic design of MMICs the accurate broadband characterisation of transmission-line characteristics, discontinuities and components is of great importance. Also, qualified circuit modelling provides the possibility for the elimination of cut-and-try methods from hybrid MIC design. For this reason, in the last decade a variety of research groups has started activities in order to characterise planar transmission lines and discontinuities by simple analytical models. On the other hand, meanwhile a number of rigorous methods exists for the frequency-dependent characterisation of thin-film transmission lines and a few approaches have been made to the rigorous description of components too. However, numerical methods for the solution of planar circuit electromagnetic boundary-value problems are too laborious and time-consuming for direct application in CAD programs. Nevertheless, they are very well suited to provide a basis for the generation of analytical approximate circuit models which is the strategy followed by the authors.

Results with a broad range of applicability can be achieved if in addition the correct asymptotic behaviour of the

considered structures for very small and very large values of the physical parameters is taken into account. Finally, for tolerance analyses and for statistical CAD concepts to be applied along with future MMIC mass production it is thought to be useful to focus some attention on the modelling of the first derivative of electrical characteristics with respect to the geometrical parameters. The accuracy which has to be achieved in modelling is mainly determined by the fabrication tolerances of thin-film structures and the variation of the parameters of the employed substrates. With highly sophisticated technologies tolerances in the order of a fraction of a percent are feasible. Therefore, approximate analytical models to be developed should exhibit an error which is about 1% or less compared to rigorous numerical solutions. Among these presuppositions the generation of models suited for MMIC CAD has been started by the authors. Analytical results have already been given for the open-end effect of microstrip,²⁻⁴ and for the asymmetric microstrip gap,⁵⁻⁷ and are presented here for the frequency-dependent effective dielectric constant of microstrip.

Results: The dispersion formula given here has been developed using Jansen's numerical hybrid mode technique^{8,9} and the associated computer programs¹⁰ as a source for the generation of a reliable data basis. As Kuester and Chang pointed out in a recent paper¹¹ this technique appears to be one of the most accurate considered by them. Furthermore, a large number of test measurements with resonating microstrip transmission-line structures on different commercially available substrates has been performed at ATE (see address) and related laboratories in the last few years and has led to agreement with the computer results¹⁰ which generally was of the order of 1% for strip widths in the whole range of technologically meaningful values.

That the analytical formulas available still need improvement can be seen by comparison with rigorous results produced according to Reference 9 and 10. This comparison is performed in Fig. 1 for the expression published by Yamashita *et al.*¹² and the more recent formula provided by Hammerstad and Jensen.¹³ The considered substrate thickness is $h = 1$ mm, the strip width is $w = h$ and the range of operating frequencies is $f = 0, \dots, 28$ GHz. It can be seen from Fig. 1 that only near $f = 0$ is an accuracy of 0.2 percent, as claimed in Reference 13, verified. Otherwise, the maximum discrepancies visible in Fig. 1 are 2.8% (Reference 12, $\epsilon_r = 12.9$, $f = 14$ GHz) and 5.4% (Reference 13, $\epsilon_r = 12.9$, $f = 26$ GHz) and are not tolerable before the background discussed here.

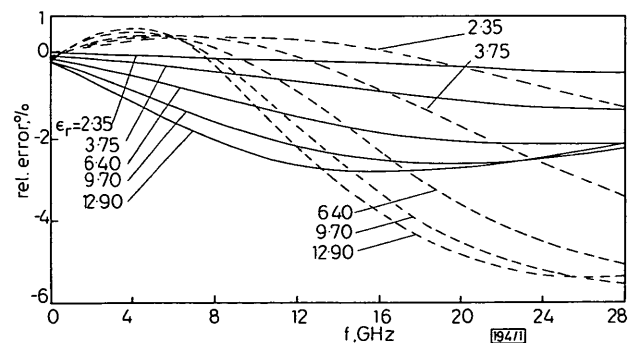


Fig. 1 Comparison between the approximate models of Yamashita *et al.*¹² and Hammerstad and Jensen¹³ and the numerical data^{9,10}

$w/h = 1.0$, $h = 1$ mm
— Yamashita *et al.*
--- Hammerstad and Jensen

The expression developed in this letter has a mathematical structure which is identical to that of Getsinger's formula describing microstrip dispersion,¹⁴ namely

$$\epsilon_{eff}(f) = \epsilon_r - \frac{\epsilon_r - \epsilon_{eff}(f=0)}{1 + P(f)} \quad (1)$$

The frequency-dependent term $P(f)$, however, has been modelled here anew, making use of the ideas mentioned in the introduction, incorporating the correct asymptotic behaviour and by computer-matching with the frequency-dependent

hybrid mode results supplied by the programs of Reference 10. With $u = w/h$ denoting the microstrip width w normalised with respect to the substrate thickness h , and fh representing the normalised frequency in GHz cm (note: $fh \sim h/\lambda_0$ where λ_0 is the free space wavelength) this results in

$$P(f) = P_1 P_2 [(0.1844 + P_3 P_4) 10 fh]^{1.5763}$$

with

$$\begin{aligned} P_1 &= 0.27488 + [0.6315 + 0.525/(1 + 0.157fh)^{20}]u \\ &\quad - 0.065683 \exp(-8.7513u) \\ P_2 &= 0.33622[1 - \exp(-0.03442\epsilon_r)] \\ P_3 &= 0.0363 \exp(-4.6u) \\ &\quad \times \{1 - \exp[-(fh/3.87)^{4.97}]\} \\ P_4 &= 1 + 2.751\{1 - \exp[-(\epsilon_r/15.916)^8]\} \end{aligned} \quad (2)$$

Compared to the converged numerical results of the employed computer program,¹⁰ the accuracy of expr. 1 in conjunction with eqns. 2 is better than 0.6% in the range $0.1 \leq w/h \leq 100$, $1 \leq \epsilon_r \leq 20$ and $0 \leq h/\lambda_0 \leq 0.13$, i.e. up to about 60 GHz for 25 mm substrates. This covers and even goes beyond useful upper operating frequencies. The range of applicability of eqns. 1 and 2 exceeds that of existing formulas considerably. For two different frequencies, $f = 10$ GHz and $f = 30$ GHz, Figs. 2a and b show the relative error $(\epsilon_{eff,num} - \epsilon_{eff,model})/\epsilon_{eff,num}$ as a function of the strip width w/h . The high accuracy of the new expression is further illuminated by Fig. 3 which depicts the first derivative of the effective dielectric constant with respect to the strip width. Thus, the agreement between the analytical model (eqns. 1 and 2) and the rigorous numerical values derived from References 9 and 10 can be seen to be in accordance with the requirements outlined before.

Applications: The new expression has been developed for use in CAD programs which includes analysis and design, tolerance analysis and the statistical design of MICs and

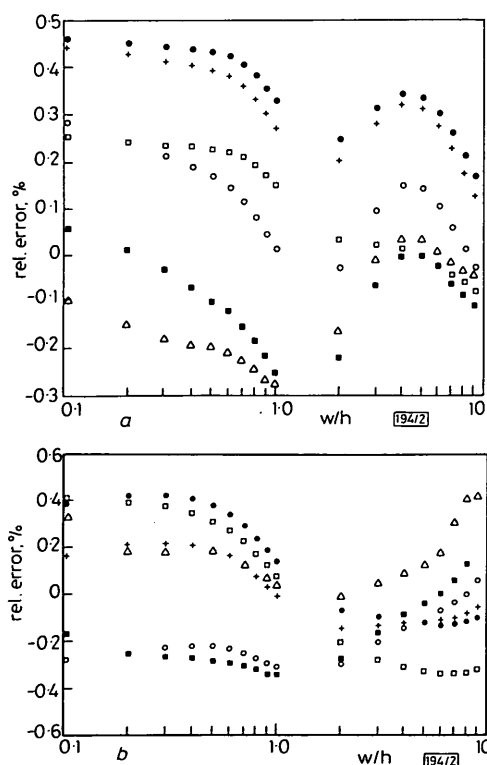


Fig. 2 Comparison between formula presented here and numerical data^{9,10}

$h = 1$ mm
 $f = 10$ GHz
 $f = 30$ GHz

□□□ $\epsilon_r = 2$ ●●● $\epsilon_r = 4$ +++ $\epsilon_r = 5$
○○○ $\epsilon_r = 8$ ■■■ $\epsilon_r = 12$ △△△ $\epsilon_r = 20$

MMICs. In addition, it is intended to use this formula together with the open-end effect expression of Reference 2 and the asymmetric gap model of Reference 5 in the measurement of microwave substrate dielectric constants by strip or ring resonator techniques. Using the presented model for the simulation of the results obtained by the mentioned measurement techniques it should be possible to determine substrate dielectric constants ϵ_r with accuracies in the order of the maximum errors inherent in $\epsilon_{eff}(f)$ and without time-consuming computer programs.

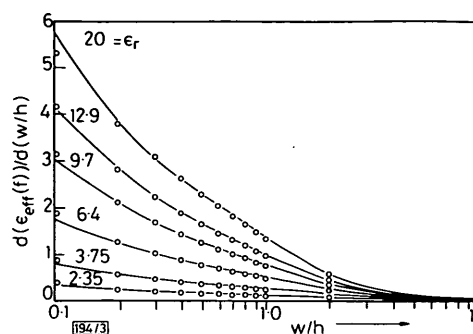


Fig. 3 Comparison of derivative $d(\epsilon_{eff}(f))/d(w/h)$ as following from the model presented and the rigorous numerical solution^{9,10}

$f = 30$ GHz, $h = 1$ mm
○○○ References 9 and 10
— analytical model

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